Assignment No: 6

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Code:

```
# Linear Regression using Boston Housing Dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt


from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score


# ------------------------------------------------
# a. Data Preparation and Preprocessing
# ------------------------------------------------

# Load dataset
data = pd.read_csv("Boston (1).csv")

print("\nDataset Shape:")
print(data.shape)

# Remove unnecessary column
if "Unnamed: 0" in data.columns:
    data = data.drop("Unnamed: 0", axis=1)

# Check missing values
print("\nMissing Values:")
print(data.isnull().sum())


# Separate features and target
X = data.drop("medv", axis=1)
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```
y = data["medv"]

# Imputation of missing values using mean
imputer = SimpleImputer(strategy="mean")

X = imputer.fit_transform(X)

print("\nMissing values after imputation:")
print(pd.DataFrame(X).isnull().sum().sum())

# Randomly split data into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2,random_state=42
)

print("\nTraining Data:", X_train.shape)
print("Testing Data:", X_test.shape)


# ------------------------------------------------
# b. Build Linear Regression Model
# ------------------------------------------------

model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Predict values
y_pred = model.predict(X_test)


# ------------------------------------------------
# c. Evaluate Model Performance
# ------------------------------------------------

# Mean Squared Error
mse = mean_squared_error(y_test, y_pred)

# Root Mean Squared Error
rmse = np.sqrt(mse)

# R2 Score
r2 = r2_score(y_test, y_pred)
```

```
print("\nModel Performance")
print("-------------------------")
print("Mean Squared Error (MSE):", mse)
print("Root Mean Squared Error (RMSE):", rmse)
print("R2 Score:", r2)


# ------------------------------------------------
# Actual vs Predicted Values
# ------------------------------------------------

result = pd.DataFrame({
    "Actual Price": y_test.values,
    "Predicted Price": y_pred
})

print("\nActual vs Predicted:")
print(result.head(10))


# ------------------------------------------------
# d. Analyze and Interpret Results
# ------------------------------------------------

print("\nAnalysis and Interpretation")
print("-------------------------")

print("MSE represents the average squared difference")
print("between actual and predicted house prices.")

print("\nRMSE represents the typical prediction error")
print("in the same units as the target variable.")

print("\nR2 Score represents the proportion of variation")
print("in house prices explained by the model.")

if r2 >= 0.7:
    print("\nThe model has a good fit.")
elif r2 >= 0.5:
    print("\nThe model has a moderate fit.")
else:
    print("\nThe model has a weak fit.")
```

```
# ------------------------------------------------
# e.Graph: Actual vs Predicted House Prices
# ------------------------------------------------

plt.figure(figsize=(8, 5))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual House Prices")
plt.ylabel("Predicted House Prices")
plt.title("Actual vs Predicted House Prices")

# Perfect prediction line
plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()]
)

plt.show()
```

Output:

```
Dataset Shape:
(506, 15)

Missing Values:
crim       0
zn         0
indus      0
chas       0
nox        0
rm         0
age        0
dis        0
rad        0
tax        0
ptratio    0
black      0
lstat      0
medv       0
dtype: int64

Missing values after imputation:
0

Training Data: (404, 13)
Testing Data: (102, 13)

Model Performance
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Mean Squared Error (MSE): 21.61768594357564
Root Mean Squared Error (RMSE): 4.649482330709048
R2 Score: 0.7212204000246899

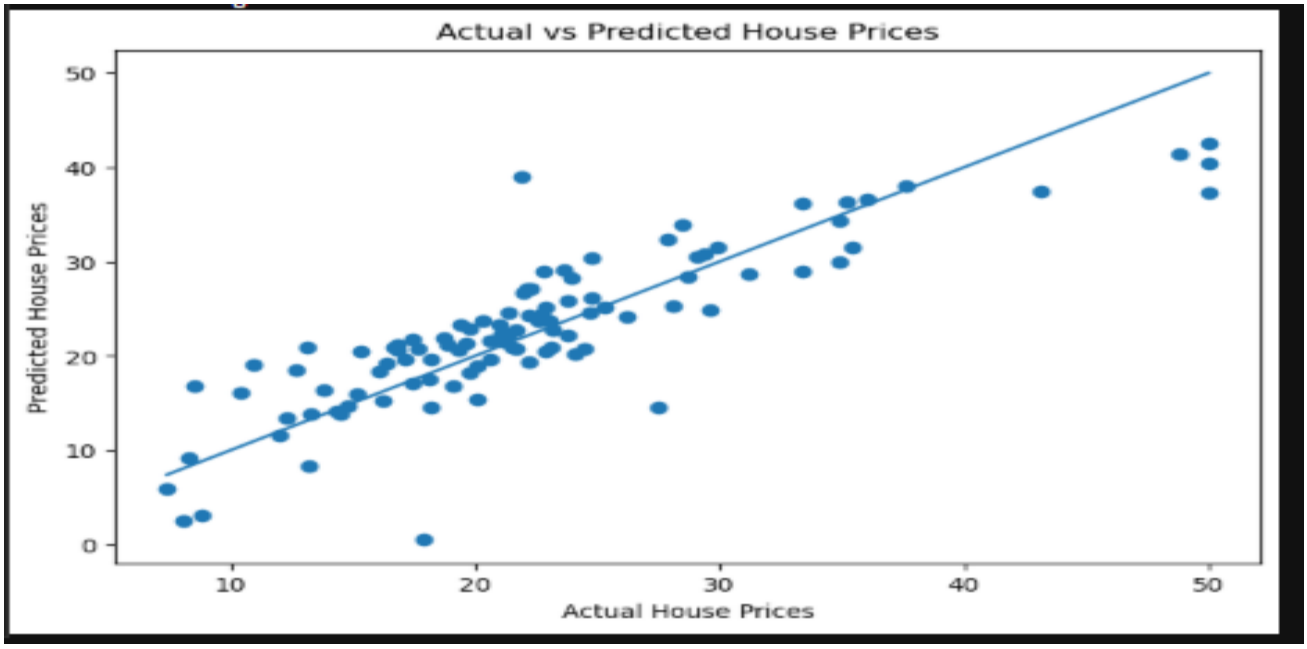
Actual vs Predicted:
   Actual Price  Predicted Price
0          19.1        16.876075
1          25.3        25.166051
2          35.2        36.380108
3          31.2        28.670450
4          22.9        20.451400
5          23.8        25.835410
6          21.5        20.975803
7          24.5        20.713578
8          20.6        21.620377
9          50.0        40.350655

Analysis and Interpretation
-------------------------
MSE represents the average squared difference
between actual and predicted house prices.

RMSE represents the typical prediction error
in the same units as the target variable.

R2 Score represents the proportion of variation
in house prices explained by the model.

The model has a good fit.
```


Actual vs Predicted House Prices
Predicted House Prices
Actual House Prices
0
10
20
30
40
50
10
20
30
40
50